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SARDAR PATEL UNIVERSITY
SYBCA 4th SEMESTER CBCS (On Demand) Examination-2024
US04CBCA25 || Data Structures - II

Date: 12/07/2024 Friday

Time: ~~10:00am~~ to ~~1~~ :00pm

Maximum Marks: 70

[10]

Q.1

Answer the following questions.

- 1 _____ is a rank of hierarchy of a tree.
(a) Level (b) Degree (c) Height (d) Index
- 2 _____ is a set of disjoint trees.
(a) Path (b) Forest (c) Directed tree. (d) Height
- 3 The size of one dimensional array can be calculated by using _____ formula.
(a) $2^{d+1} + 1$ (b) $2^{d-1} + 1$ (c) $2^{d+1} - 1$ (d) $2^{d-1} - 1$
- 4 Which of the following allows a portion of memory to be shared by different types of data?
(a) Four-way list (b) Three-way list (c) Two-way list (d) One-way list
- 5 A linked list is considered as an example of _____ type of memory allocation.
(a) Dynamic (b) Static (c) Compile time (d) None of the Above
- 6 LLINK / LPTR is the pointer pointing to the _____.
(a) Successor node (b) Predecessor node (c) Head node (d) Last node
- 7 _____ is the process to find the particular element from the array.
(a) Sorting (b) Inserting (c) Searching (d) Updating
- 8 Quick sort is also known as _____.
(a) Merge sort (b) Heap sort (c) Bubble sort (d) None of these
- 9 The _____ is an area into which data records are written when the file is first created.
(a) Prime area (b) Control area (c) Index area (d) Overflow area
- 10 Overflow areas are _____ overflow area and _____ overflow area.
(a) Cylinder, Independent (b) Independent, Track (c) Track, Cylinder (d) Prime, Track

[20]

Q.2

Explain following in brief. (Attempt any Ten)

- 1 What is Binary Tree? Also give its example.
- 2 List out Applications of Tree.
- 3 Draw the Binary Tree for $(A-B)+C*(E/F)$
- 4 Explain linked list in brief.
- 5 Explain doubly circular linked list in brief.
- 6 List delete operations that can be performed on singly linked list.
- 7 List out application of sorting.
- 8 Define: Internal Sort and External Sort.
- 9 Define searching. List out the names of searching techniques.
- 10 Define Bucket Capacity.
- 11 Define: Hashing function, Collision resolution technique
- 12 Explain in brief prime area.

[P.T.O.]

- Q-3 (A) Explain the insertion of a node in a binary with an example. [05]
(B) Draw a binary tree for the expression 1. $A*B-(C+D)*(P/Q)$ 2. $A+(B-C)*(E*F)/(G/I)$. [05]

OR

- Q-3 (A) Explain deletion of a node from binary tree (all 3 conditions) with example. [05]
(B) Write down an algorithm for postorder traversal of a binary tree. [05]

- Q-4 (A) List and explain types of linked lists with example. [05]
(B) Write an algorithm to delete node at beginning in singly linked list. [05]

OR

- Q-4 (A) Write an algorithm to delete node at end position in singly linked list. [05]
(B) Write an algorithm to insert node at front position in singly linked list. [05]

- Q-5 (A) Write down the algorithm of bubble sort. [05]
(B) Write down the algorithm of Linear Search. [05]

OR

- Q-5 (A) Write down the algorithm of Binary Search. [05]
(B) Write down the algorithm of Merge sort. [05]

- Q-6 Write a detail note on structure and processing of Direct files. [10]

OR

- Q-6 List and explain popular hashing function with example. [10]

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